

Logical operations:

→ A logical operators allow you to test for the truth of a condition. ~~Similar~~.

→ The following table illustrates the SQL logical operators.

Operator	meaning.
ALL	* Return true if all comparisons are true
AND	* Return true if both expressions are true.
ANY	* Return true if any one of the comparison is true
BETWEEN	* Return true if the operand is within a range.
Equals IN	* Return true if the operand is equal to one of the value in a list
Is EXISTS	* Return true if the sub query contains any rows

1. AND: The AND operator allow you to construct multiple conditions in the WHERE clause of an SQL statement such as select.

→ The following example finds all employees whose salaries are greater than 5000 and less than 7000

> Select first-name, last-name, salary from employees where salary ≥ 5000 ¹⁸
AND salary ≤ 7000 order by salary;

output:

first-name	last-name	salary
John	Wesly	6000
Elen	daniel	6000
Luies	Popp	6900
Shanta	Suji	6500

2. ALL: The ALL operator compares a value to all values in another value set.

→ The following example finds all employees whose salaries are greater than all salaries of employees

> Select first-name, last-name, salary from employees where
salary $\geq$ ALL (select salary from employees
where department_id = 8) order by salary
DESC;

output:

first-name	last-name	Salary
Steven	King	24000
John	Russel	17000
Neena	Kochhar	14000

3. ANY: The ANY operator compares a value to any value in a set according to condition.

The following example statement finds all employees whose salaries are greater than the average salary of every department.

> Select first-name, last-name, salary from employees where
 salary > ANY (Select Avg(salary) from employees
 group by department-id) order by first-name,
 last-name;

Output:

first-name	last-name	salary
Alexander	Hunold	9000.00
Charles	Johnson	6200.00
David	Austin	4800.00
Den Adam	Flip	9000.00

4. Between: the between operator searches for values that are
 within a set of values.

→ For example, the following statement finds all employees
 whose salaries are between 9000 and 12000.

> Select first-name, last-name, salary from employees where
 salary between 9000 AND 12000
 order by salary;

Output:

first-name	last-name	salary
Alexander	Hunold	9000.00
Den	Reichards	10000.00
Nancy	Prince	12000.00

5. IN: The IN operator compares a value to a list of specified values. The IN operator returns true if the compared value matches at least one value in the list.

→ The following statement finds all employees who work in the department id 8 or 9.

> select first-name, last-name, department-id from employees
where department-id IN (8,9) order by
department-id;

output:

First-name	last-name	Department-id.
John	Russel.	8
Jack	Livingstone	8
Steven	King	9
Neena	Kochhar	9

6. EXISTS:- The EXISTS operator tests if a subquery contains any rows

→ For example, the following statement finds all employees who have dependents

> select first-name, last-name from employees e where EXISTS
(select 1 from dependent d where
d.employee-id = e.employee-id);

First-name	last-name
Steven	King
Neena	Kochhar
Alexander	Hunold.